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Transoral single-port robotic surgery for benign or early stage malignant tumors of pharynx and larynx - a prospective real-world study from mainland China

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Abstract

Background Transoral robotic surgery (TORS) is a minimally invasive approach for treating early-stage malignant or benign tumors of the pharynx and larynx. This study represents the first real-world, single-arm, early feasibility evaluation of clinical performance and perioperative safety of da Vinci Single-Port (SP) for treating various benign or early-stage malignant tumors of the pharynx and larynx in Mainland Chinese population.

Method A real-world, single-arm, single-center, early feasibility study (ChiCTR2300078809) enrolled 15 patients with a heterogeneous benign or early-stage malignant tumors of pharynx and larynx, who received da Vinci single port-assisted transoral robotic surgery (SP-TORS) from July 2023 to May 2024. The primary endpoint was intraoperative conversion rate, and the secondary endpoints included intraoperative bleeding, surgical duration, length of stay (LOS), ICU admission, postoperative pain score (VAS) and recovery of swallowing functions (MDADI).

Results All 15 patients successfully completed SP-TORS without intraoperative conversion, ICU admission, intraoperative or postoperative massive bleeding. The median intraoperative bleeding volume and surgical duration were 5 ml and 60 min, respectively. The median LOS was 4 days with relatively low postoperative VAS. MDADI indicated all patients significantly recovered swallowing functions 1 month after surgery.

Conclusion Our preliminary institutional experience demonstrated clinical feasibility and acceptable perioperative safety for treating early-stage malignant or benign tumors of the pharynx and larynx in the Mainland Chinese population through SP-assisted transoral robotic surgeries.

Keywords Single-port robot · TORS · Oropharynx · Hypopharynx · Larynx

Introduction

Transoral Robotic Surgery (TORS) is a surgical technique that utilizes robotic arms to resect tumors in the pharynx, larynx, and adjacent areas through an oral approach. In 2005, Weinstein et al. [1] pioneered the use of TORS by performing a supraglottic laryngectomy in a canine model, demonstrating its potential application in otorhinolaryngology and head and neck surgery. Since then, numerous animal and clinical studies have validated the safety and oncological efficacy of TORS [2]. Currently, common transoral minimally invasive surgical approaches for treating benign and malignant tumors of the pharynx and larynx include transoral laser microsurgery (TLM) [3], plasma surgery, and multi-arm robotic-assisted surgery. TORS has

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been widely adopted for the surgical management of various pharyngeal and laryngeal lesions, such as oropharyngeal cancer [4], laryngopharyngeal cancer [5], supraglottic laryngeal cancer [6], parapharyngeal space tumors [7], and intraoral thyroglossal duct cysts [8]. Numerous studies have demonstrated that, compared to open surgeries, TORS offers significant advantages, including reduced surgical trauma, fewer complications, and comparable oncological outcomes [8]. Despite the proven benefits of multi-arm robotic-assisted TORS, its application in transoral surgeries is often limited by challenges such as difficulty in placement within confined spaces, frequent instrument collisions, and the "chopstick effect," where robotic arms interfere with each other during operation [9]. These limitations highlight the need for more advanced robotic systems tailored to the unique demands of transoral procedures.

The Da Vinci SP surgical system, launched in 2014, represents the latest advancement in single-port robotic surgery. It features an endoscope and three robotic arms deployed through a single 2.5 cm round port, capable of rotating 270°. Its double-jointed, flexible robotic arms offer superior maneuverability compared to multi-arm systems. The SP system is particularly well-suited for narrow anatomical spaces, enabling easier docking, better exposure, and more efficient operation during surgery. These features make it an ideal choice for transoral procedures such as TORS. However, to date, there are only a few reported cases of the SP system being used in oropharyngeal and laryngopharyngeal surgeries [10–13], and no studies have been conducted in Mainland China. Therefore, we conducted a real-world, single-arm, early feasibility study to evaluate the clinical performance and perioperative safety of the SP system, exploring the efficacy and perioperative safety in transoral surgery for various tumors of the oropharynx, hypopharynx, and larynx, providing real-world evidence for its early feasibility application in Chinese population.

Materials and methods

Study design

The SP system is not officially launched in China yet. However, in accordance with the *Regulations on the Management of Imported Medical Devices Urgently Needed for Clinical Use in the Hainan Free Trade Port Boao Lecheng International Medical Tourism Pilot Zone*, the SP system was introduced to Ruijin-Hainan Hospital Affiliated to Shanghai Jiao Tong University School of Medicine (Boao Research Hospital) for early feasibility clinical application. Following ethical approval (KY2023 LS No. 002), a prospective, single-center, single-arm, real-world, early feasibility study (ChiCTR2300078809) was conducted. This early feasibility study enrolled 15 consecutive subjects who met inclusion and exclusion criteria and voluntarily opted for SP robotic surgery to treat benign lesions or T1/T2 malignant tumors in the oropharynx, hypopharynx and larynx. The sample size is non-powered due to the nature of early feasibility study.

Inclusion criteria were: (1) Age ≥ 18 years; (2) Patients who plan to receive surgical treatment for benign lesions or T1/T2 malignant tumors in the oropharynx, hypopharynx and larynx using the SP system; and (3) Patients who voluntarily decide to participate in the study and provided signed informed consent (ICF). Exclusion criteria were: (1) Patients with contraindications for single-port robotic surgery; (2) Patients with missing data on the primary endpoint in retrospective cases; and (3) Patients who are considered inappropriate to participate in this study by the investigators. All subjects were recruited at the Department of Otolaryngology Head and Neck Surgery of Ruijin Hospital Affiliated to Shanghai Jiao Tong University School of Medicine from July 2023 to May 2024, and received surgeries at Ruijin-Hainan Hospital due to the Boao policy.

The primary study endpoint was the intraoperative conversion rate, and the secondary study endpoints included intraoperative bleeding volume, surgical duration, length of stay (LOS), admission to ICU and ICU LOS, pathologically confirmed positive surgical margin, postoperative pain score (VAS), and recovery of eating and swallowing functions after surgery (MDADI).

Preoperative preparation and instruments used with the SP system

Trans-nasal anesthetic intubation was performed on the contralateral side of the patient's lesion maximize space at the surgical site. Following the induction general anesthesia, the patient was placed in a supine position, the endotracheal tube was securely fixed to prevent dislodgement, and the surgical area was routinely disinfected. Da Vinci SP Surgical System (model: SP1098) was used for all cases. The SP System Cart was generally placed at the upper left of the patient's head. The surgeon sat in front of the console, and the assistant sat either on the left or procedure side of the patient's head to assist with sucking out secretions, replacing or wiping endoscope and adjusting for better surgical field exposure (Fig. 1) through the SP display screen. As for da Vinci SP instruments, the most frequently used were Maryland Bipolar Forceps, Cautery Spatula, and Monopolar Curved Scissors.

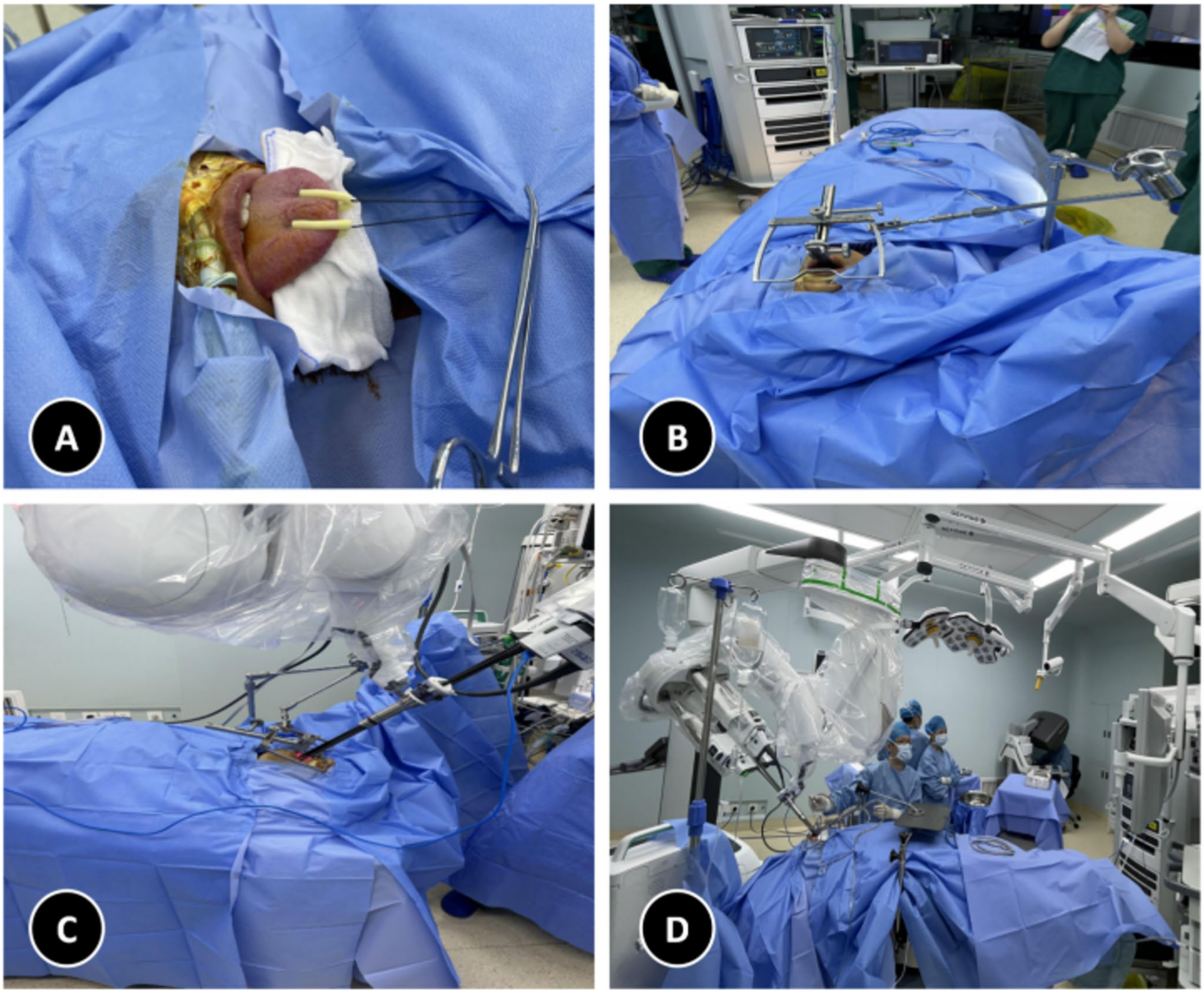

Fig. 1 The SP robot was prepared for TORS. **A** The tongue body was pulled to better expose the tumor in the tongue base, larynx, or hypopharynx; **B** A mouth-gag was placed; **C** The robot was docked; **D** TORS was performed

Surgical techniques

Tonsillar tumor Under endoscopic visualization, the tumor was resected en bloc with a 0.5 cm margin from the tumor boundary using a Cautery Spatula. During operation, if the lesion was on the right side of the patient, a pair of Maryland Bipolar Forceps was held using the left hand for pulling the tissue, and a Cautery Spatula or Monopolar Curved Scissors was held using the right hand for cutting. If the lesion was on the left side of the patient, then switch the instruments in two hands. Electrocoagulation was generally used for cutting, at an energy intensity level of 3 or 4. In case of significant bleeding, a pair of Bipolar Coagulation Forceps was used for hemostasis. In case when fat in the parapharyngeal space was exposed or most of the superior constrictor muscle of the pharynx was resected, a pedicled mucosal flap was utilized to cover on the posterior pharyngeal wall.

Tumors in tongue base, larynx, and hypopharynx To achieve optimal tumor exposure, the tongue body should be pulled out of the mouth. A 0–0 silk thread was sewed 0.5 cm lateral to the midline of the tongue and 2 cm away from the tongue tip respectively to pull out the tongue. A wet gauze pad was positioned on the lower incisors to prevent injury to the tongue belly. A mouth-gag was then inserted to fully expose the tumor. The lesion exposure was observed under endoscope until satisfactory achieved. A Cautery Spatula was used to mark the excision line 1 cm away from the tumor boundary by spot electric coagulation, and then the tumor was resected en bloc. The instrument handling method was the same as tonsillectomy.

Patient were followed up in the outpatient clinic 1 month later to observe wound recovery (Fig. 2).

Statistical method

SPSS 24.0 was used for statistical analysis. All data are presented as median (range). For short-term postoperative pain and recovery comparison, paired t-test was used for data that satisfies normal distribution, such as postoperative pain and recovery indicators before and after surgery. Wilcoxon rank sum test was used for data in which the normal distribution assumption could not be assured. In all cases, statistical significance was achieved at $P<0.05$.

Results

Patient baseline characteristics

A total of 15 patients were enrolled, including 10 males and 5 females, with a median age of 50 (31–78) years old. Among the 15 patients, 5 were diagnosed with benign lesions (2 epiglottic cyst, 1 inflammatory hyperplasia of lymphoid tissue at tongue base, 1 chronic tonsillitis, and 1 intraoral thyroglossal duct cyst). The remaining 10 patients had malignant lesions (2 supraglottic squamous carcinoma, 5 tongue base squamous carcinoma, 2 tonsillar squamous carcinoma, and 1 hypopharyngeal squamous carcinoma) (Table 1). Of the 7 cases with oropharyngeal squamous carcinoma, 6 tested positive for HPV16 (Table 2). For primary tumor staging prior to SP surgery, one patient was classified with T1 stage, 6 as T2 and 3 as T3. All T3 patients underwent two courses of preoperative chemotherapy, resulting in tumor downstaging to T1 in 2 patients and T2 in 1 patient before surgery and thus all patients met the inclusion criteria

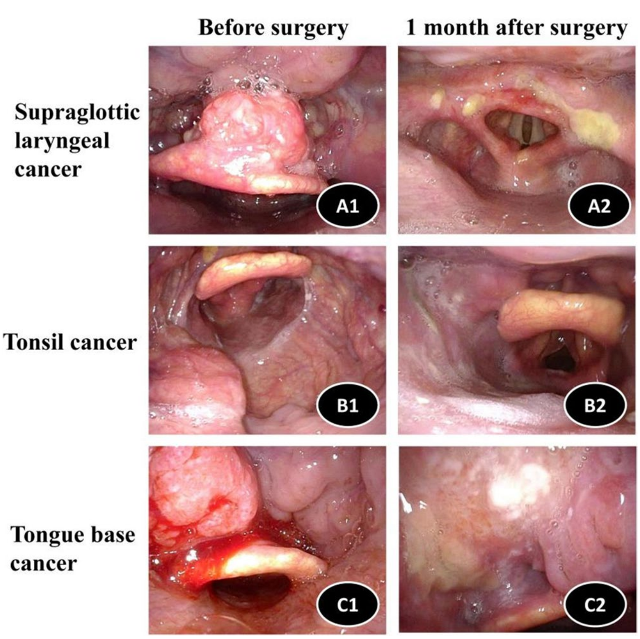

Fig. 2 Laryngoscopy of patients with cancers at different sites before surgery and 1 month after surgery. A1-2: Laryngoscopy images of a patient with supraglottic laryngeal cancer before surgery and 1 month after surgery. B1-2: Laryngoscopy images of a patient with tonsil cancer (left side) before surgery and 1 month after surgery. C1-2: Laryngoscopy images of a patient with tongue base cancer (left side) before surgery and 1 month after surgery

Table 1 Patient Baseline Characteristics

Characteristics		Total (*n*)
Gender	Male	10
	Female	5
Age	[median, (range)], year-old	50 (31–78)
Tumor Site	Tonsil	3
	Tongue Base	7
	Larynx	4
	Hypopharynx	1
Pathological Type		
Benign	Chronic Tonsillitis	1
	Thyroglossal Cyst	1
	Epiglottic Cyst	2
	Lymphadenosis at Tongue Base	1
Malignant	Supraglottic Squamous Carcinoma	2
	Tongue Base Squamous Carcinoma	5
	Tonsillar Squamous Carcinoma	2
	Hypopharyngeal Squamous Carcinoma	1

Table 2 Information on Patients with Malignant Tumors

Characteristics		Oropharynx (n=7)	Larynx (n=2)	Hypopharynx (n=1)
Pathological Tumor Stage				
T	T1	0	2	0
	T2	7	0	1
N	N0	1	1	1
	N1	0	0	0
	N2a	0	0	0
	N2b	5	0	0
	N2c	0	1	0
	N3	1	0	0
M	M0	7	1	1
	M1	0	1	0
HPV16	+	6	0	0
	-	1	2	1
Clinical Stage	II	1	1	1
	III	6	0	0
	IV	0	1	0
Degrees of Differentiation				
Well Differentiated		1	0	0
Moderately Differentiated		2	2	1
Poorly Differentiated		4	0	0

of this study. For cervical lymph node metastasis, 3 patients were diagnosed with N0 stage (2 oropharyngeal carcinoma, 1 laryngeal carcinoma), 6 patient were N2 (5 oropharyngeal carcinoma, 1 laryngeal carcinoma), and 1 patient was N3 (oropharyngeal carcinoma) cases. 6 patients with N2 and 1 patient with N1 received additional selective or radical neck dissection before or after SP TORS.

SP TORS surgical outcomes

All SP-TORS procedures were completed without any intraoperative conversion, and all patients were transferred to general wards after surgery with no ICU admission needed (Table 3). The total surgical duration (measured from the first satisfactory lesion exposure to the completion of surgery) ranged from 7 to 186 min, with a median of 60 min. The console duration (for tumor resection) ranged from 2 to 130 min, with a median of 22 min. Subgroup analysis showed the median surgical durations for malignant and begins cases were 66 (50–186) min and 48 (7–70) min, respectively. The median console durations for malignant and begins cases were 23.5 (5-130) min and 15 (2–27) min, respectively. Overall, 11 cases (7 malignant and 4 benign) were completed within 30 min of console time.

The intraoperative bleeding volume ranged from 0 to 50 ml with a median of 5 ml, and none of the patients required intraoperative blood transfusion. Subgroup analysis showed the median bleeding volumes for malignant and begin cases were 5 ml (1–50 ml) and 5 ml (0–15 ml), respectively. All malignant patients had pathological examination of resected specimens, and one case of tongue base carcinoma (1/10) had a positive surgical margin. In patients with malignant tumors, oropharyngeal carcinoma accounted for the largest proportion (7/10), with a HPV16 positive rate of 85.7% (6/7) (Table 2), which is basically consistent with the incidence of HPV+oropharyngeal cancer in the Europe and America [14].

As for complications, no SAE occurred intraoperatively or postoperatively. Three patients experienced surgery-related AEs (3/15), all of which were malignant cases. Specifically, one patient with tongue base carcinoma had loose teeth (CTCAE Grade 1), one case with supraglottic laryngeal carcinoma had lost teeth (CTCAE Grade 1), and another case with tongue base carcinoma experienced laryngeal obstruction within 2 h post-surgery (CTCAE Grade 3) and received an immediate tracheotomy (Table 3).

All 15 patients resumed oral intake and eating functions within 72 h after surgery. Among them, 14 patients were able to tolerate a soft oral diet and regained eating function within 6 h postoperatively, with no significant swallowing difficulties observed. One patient with supraglottic laryngeal carcinoma required a nasogastric tube inserted post-surgery, which was removed after 3 days of swallowing exercises.

The median LOS was 4 days (2–11 days) for all patients. Subgroup analysis showed for malignant and benign cases, the median LOS was 4 days (2–11 days) and 2 days (2–5 days), respectively (Table 3).

Postoperative pain and recovery

Patients were asked to score their pain intensities at 1 day and 3 days after surgery, using the VAS scoring system. The mean VAS score of all patients was 3.07 ± 1.91 one day after surgery, and 2.2 ± 1.2 three days after surgery, which showed a relief of postoperative pain (Table 3).

Table 3 Intraoperative and Postoperative Outcomes

Characteristics	Total
Intraoperative Conversion Rate (95% CI), %	0 (0.0， 21.8)
Transfer to ICU, *n*	0
Surgical Duration [mean ± SD, (range)], minutes	73.1 ± 48.4 (7–186)
Malignant	86.2 ± 52.9 (50–186)
Benign	46.8 ± 24.6 (7–70)
Console Duration [mean ± SD, (range)], minutes*	34.5 ± 48.4 (2–130)
Malignant	44.6 ± 47.0 (5–130)
Benign	14.4 ± 24.6 (2–27)
Bleeding Volume [mean ± SD, (range)], ml	9.0 ± 12.1 (0–50)
Malignant	10.2 ± 14.4 (1–50)
Benign	6.6 ± 5.9 (0–15)
Positive Surgical Margin, *n*	1
LOS [mean ± SD, (range)], days	4.1 ± 2.3 (2–11)
Malignant	4.8 ± 2.4 (2–11)
Benign	2.8 ± 1.3 (2–5)
AEs, *n*	3
Loose/Lost Teeth	2 (CTCAE 1)
Tracheotomy	1 (CTCAE 3)
Postoperative Gastric Tube Indwelling, *n*	1
Follow-up Duration [median ± SD, (range)], months	13.5 ± 6.2 (6–17)
Postoperative pain score (VAS) [mean ± SD]	
1 Day After Surgery	3.1 ± 1.9
3 Days After Surgery	2.2 ± 1.2
P Value†	0.144
MDADI [mean ± SD]	
1 Day Before Discharge	61.9 ± 11.1
1 Month After Surgery	78.2 ± 15.9
P value‡	0.001

*Duration of robot operation by the surgeon at the console

†Comparison of VAS-based scores 1 day and 3 days after surgery

‡Comparison of MDAI-based scores 1 day before discharge and 1 month after surgery

As mentioned above, only 1 case with supraglottic laryngeal carcinoma (1/15) received nasogastric tube insertion after surgery. The nasogastric tube was inserted to prevent the patient from bleeding caused by choking and coughing while eating, the tube was then removed after no obvious choking and coughing were found when oral diet was recovered 3 days after surgery.

In this study, we also compared the recovery of patients' swallowing functions at 1 day before discharge and 1 month after discharge, using the MDADI. The mean MDADI score was 61.9 ± 11.11 at one day before discharge, and 78.2 ± 15.95 at one month after discharge, indicating improvement in the swallowing functions (Table 3).

Postoperative follow up

Patients with benign tumors received following up 1 month after surgeries. Following up contents including physical examination and laryngoscopy. Patients with malignant tumors received following up every month in the first 6 months and every 3 months after 6 months within 2 years according to NCCN guidelines [15]. Following up contents including physical examination and laryngoscopy each time. And patients received ultrasounds, CT/MRI scan every 9 months. For suspicious lesions, further biopsies will be performed. Only 1 out of the 10 malignant patients had a positive surgical margin reported, who was a patient with tongue base carcinoma. Among the 10 patients diagnosed with malignant tumors, 8 received adjuvant radio-chemotherapy after surgery, and 7 with positive cervical lymph node metastasis underwent additional selective or radical neck dissection either before or after SP TORS. The follow-up durations of these 10 patients all exceeded 6 months, with 6 patients exceeding 12 months. The median follow-up duration was 13.5 months (6–17 months). Each patient was followed up every month in the first 6 months and every 3 months after 6 months. The examinations include laryngoscopy each time and CT or MRI every 6 momths. There were no local recurrence, death or distant metastasis cases. Only one patient with tongue base carcinoma (T2N2aM0) refused to receive radio-chemotherapy after TORS and unilateral neck dissection, and developed contralateral cervical lymph node metastasis 11 months after surgery.

Discussion

Similar to multi-arm robotic-assisted TORS, SP system has been commonly used for oropharyngeal [13, 16] and supraglottic laryngeal cancer [10, 17], but there is limited data in the treatment of hypopharyngeal, glottic laryngeal and nasopharyngeal cancers. In addition, reports on da Vinci SP TORS application mainly come from Korea [18, 19], America [20], Italy [7], Poland [10] and Hong Kong [13, 21]; there is a conspicuous absence of reports on the application of SP-TORS within Mainland China considering many factors such as scope of application, cost, medical insurance policy. But the incidence of head and neck squamous cell carcinoma ranked 7th in China, and the 5-year survival rate only ranges from 40% to 50% with little improvement in recent years [22]. Also, limited by the narrow exposure of traditional minimally invasive transoral surgery, open surgeries usually lead to larger trauma and effect the quality of life after surgeries,. Thus, it is urgently needed to introduce da Vinci SP system. We then conducted a real-world, single-arm, early feasibility study to evaluate the clinical performance and perioperative safety of the SP system for transoral otolaryngology surgeries in Mainland China. A cohort of 15 subjects with a variety of benign lesions or early stage malignant tumors in the oropharynx, hypopharynx and larynx were enrolled. Based on our preliminary institutional experience, all patients successfully completed TORS without any intraoperative conversion, transfer to ICU, postoperative massive bleeding or death, indicating the clinical feasibility and perioperative safety of da Vinci SP Surgical System in our small cohort study. However, more experience needs to be summarized after collecting more patients suitable for receiving SP-TORS.

In our early feasibility study, the median surgical duration for all cases was 60 min (7–186 min). Subgroup analysis indicated the median surgical durations for malignant and begin cases were 66 min (50–186 min) and 48 min (7–70 min), respectively. The mean surgical bleeding volume of 10 patients with malignant tumors was 10.2 ± 14.35 ml, this might be associated with the small sample size and early cancer stages. These intraoperative surgical outcomes were consistent with the previous results reported by Oberhelman [11], Costantino [9] and Park [19].

Many researchers pointed the importance of prophylactically ligating relevant branches of the external carotid artery during surgeries on tongue base and oropharynx [23]. For example, Oberhelman et al. reported that 86.4% of the patients (76/88, SP group) with oropharyngeal cancers underwent prophylactic arterial ligation during surgeries [11]. However, the cases in our early feasibility study did not receive such prophylactic operations. Based on our preliminary institutional experience, we believe that the surgeon must first have sufficient experience in transoral minimally invasive surgeries (laser and plasma). Then the preoperative contrast-enhanced CT examination is crucial to understand the relationship between blood vessels and the tumor. In addition, the lesion should be fully exposed during surgery (for surgeries on tongue base and larynx, the tongue body should be pulled outward) and a clear field of vision should be maintained. Timely use of Bipolar Coagulation Forceps to address exposed small blood vessels or active bleeding happened could help to reduce the risk of postoperative bleeding.

Regarding postoperative complications, two patients had tooth injury (one loose teeth and one lost teeth) and another patient experienced laryngeal obstruction after surgery and received tracheotomy subsequently. This complication rate is slightly higher than previously reported data from Oberhelman [11], Holsinger [12] and Stanley [24]. This could be due to the small sample size and the fact that we are still at the very early learning curve stage in real-world application, as the surgeon was only performing the very first SP-TORS cases. We anticipate that complication rates may decrease as surgical experience and volume increase, as reported by Stanley et al. In three AE cases, the two patients with tooth injury already had loosen teeth and overbite before SP procedure. Their teeth was injured during the procedure when better lesion exposure was needed. For the other AE, the patient with malignant tongue tumor developed laryngeal obstruction due to severe swelling of tongue. We believed this was caused by two factors: firstly, the tongue was being pulled outside for an extended period (176 min); and secondly, the patient underwent a radical neck dissection 2 weeks prior to SP surgery and thus could potentially affect the oropharyngeal venous return. The combination of the two reasons caused severe swelling of the tongue. This case highlights the importance of careful intraoperative management. When the tongue is pulled outward to fully expose the tumor at tongue base, epiglottic vallecula or hypopharyngeal cavity, it is crucial not only to protect the ventral surface of tongue from injury by the lower incisors but also to periodically relax the traction line and tongue depressor every 30 min during prolonged surgeries.

For postoperative pain and recovery, we found that compared to 1 day after surgery, the VAS scores at 3 days after surgery were reduced but without any significant difference. We also evaluated the MDADI score, which increased significantly at 1 month after surgery, indicating patients' swallowing functions significantly recovered. Only one patient had nasogastric tube insertion for 3 days. Oberhelman et al. reported that the duration of nasogastric tube placement was 10.4 ± 8.1 days in SP group [11], and Park et al. reported a similar time of 11 days for patients with laryngeal malignant tumor undergoing SP surgery [19]. We recommend when ensuring airway safety after surgery, patients should resume oral diet as soon as possible, which might help to reduce surgical cavity infections and relieve pain.

During postoperative follow up, we observed a negative surgical margin rate of 90%. We attribute the single positive surgical margin in our study to several factors: (1) this patient had tongue base carcinoma downgraded from T3 to T2 after chemotherapy and thus received surgery for T2; (2) the lesion was superficial, making it difficult to distinguish the papilla on the surface of the tongue base from the tumor parenchyma during surgery; (3) this was a case completed at the very initial stage during learning curve, the tongue was not fully pulled outward during surgery, potentially leading to insufficient tumor exposure. The median follow-up duration of 10 malignant cases was 13.5 months (6-17months) in our early feasibility study, and there was no death or local recurrence so far. Only one patient (with oropharyngeal cancer, T2N2bM0) was found to have contralateral lymph node metastasis at 11 months follow-up due to refusal to receive radio-chemotherapy after surgery. However, one patient with oropharyngeal cancer (T2N2bM0) developed contralateral lymph node metastasis at the 11-month follow-up. This patient had declined postoperative radio-chemotherapy, which likely contributed to the metastasis. Subsequently, the patient underwent contralateral neck dissection followed by radical radio-chemotherapy and has shown no disease progression to date. Given the small sample size and relatively short follow-up duration of this early feasibility study, no oncological conclusions can be drawn. Studies with larger sample size, longer follow-up period, and preferably multi-center experience are needed.

Conclusion

This is the first report on da Vinci SP TORS for China Mainland patients with various benign or early-stage malignant oropharynx, hypopharynx and larynx tumors. Based on our preliminary institutional experience, SP demonstrated clinical feasibility and acceptable perioperative safety for treating early-stage malignant or benign tumors of the pharynx and larynx in the Mainland Chinese population. However, it is still at an early stage. Since SP TORS is not suitable for all patients considering patients' condition and the location of tumor, surgeons need to prudently select suitable patients. Also, more training needs to be carried out and experience needs to be accumulated for tumors in different parts.

Author contribution Original draft was written by BY and YS. Material preparation, data collection and analysis were performed by HW, HH, JQ and QW. Statistics was analyzed by YL, AZ, XM and ZW. Operations were performed by MX. MX and JW also reviewed the final article. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Ethics statement The studies involved human participants and was approved by the Ethical Review Board in Ruijin Hospital affiliated to Shanghai Jiaotong University School of Medicine. The authors declared that written informed consent was obtained from all participants for the publication of this manuscript and accompanying radiological images and sequencing results.

Conflict of interest The authors of this manuscript declare no relationships with any companies whose products or services may be related to the subject matter of the article. All authors declare no potential conflict of interests.

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